

CAPITAL STRUCTURE – PART B

POINT OF INDIFFERENCE

Meaning & Basic Condition

The Point of Indifference is the level of EBIT at which two different financing plans provide the same Earnings Per Share (EPS). At this point, the company is indifferent between the two financing alternatives from the EPS point of view, because both plans produce equal EPS.

$$\text{BASIC CONDITION: } EPS_1 = EPS_2$$

Purpose of Indifference Point

- Compare two financing alternatives.
- Find the EBIT level where both plans give equal EPS.
- Decide whether equity or debt financing is more beneficial.
- Study the effect of financial leverage on EPS.
- Select a suitable capital structure at different levels of EBIT.

General Formula

$$[(EBIT - I_1)(1 - T) - PD_1] / N_1 = [(EBIT - I_2)(1 - T) - PD_2] / N_2$$

Where:

- EBIT = Earnings Before Interest and Tax
- I_1, I_2 = Interest under the two plans
- T = Tax Rate
- PD_1, PD_2 = Preference Dividend
- N_1, N_2 = Number of Equity Shares

Concept Flow-chart

Two Different Financing Plans



Compare Earnings Per Share (EPS)



Find the EBIT Level where $EPS_1 = EPS_2$



POINT OF INDIFFERENCE

POINT OF INDIFFERENCE LOGIC & INTERPRETATION

Logic Behind Indifference Point

EQUITY FINANCING

- No interest burden.
- More equity shares are issued.
- Earnings are distributed among a larger number of shares.

DEBT FINANCING

- Interest is a fixed financial charge.
- Fewer equity shares are issued.
- At higher EBIT, earnings are divided among fewer shares, which may increase EPS.

Therefore: The advantage of debt financing depends upon the level of EBIT.

Interpretation

COMPARE EBIT WITH THE INDIFFERENCE POINT



EBIT < Indifference Point	EBIT = Indifference Point	EBIT > Indifference Point
All-equity plan may provide higher EPS because debt interest reduces earnings available to equity shareholders.	Both financing plans provide the same EPS.	Debt plan may provide higher EPS because fewer shares can outweigh the fixed interest burden.

Important Point

The Indifference Point is mainly a decision-making tool for capital structure planning.

Benefit of Financial Leverage = Cost of Fixed Financial Charges

Risk Note: A financing plan should not be selected only because it gives higher EPS. Management should also consider the additional financial risk created by debt financing.

Quick Revision

Indifference Point = EBIT where $EPS_1 = EPS_2$

Above Indifference Point → Debt may be better

Below Indifference Point → Equity may be better

At Indifference Point → Both plans give equal EPS

POINT OF INDIFFERENCE – PRACTICAL

GIVEN INFORMATION & PRELIMINARY CALCULATIONS

Practical Problem

A company requires ₹20,00,000 and is considering two financing plans:

PLAN A – ALL EQUITY

Equity Capital = ₹20,00,000

PLAN B – EQUITY + DEBT

Equity Capital = ₹12,00,000

Debt = ₹8,00,000 @ 10%

Other Information

Equity Share Price

₹20

Tax Rate

25%

Step 1: Number of Equity Shares

Number of Equity Shares = Equity Capital ÷ Equity Share Price

PLAN A

₹20,00,000 ÷ ₹20 = 1,00,000 shares

PLAN B

₹12,00,000 ÷ ₹20 = 60,000 shares

Step 2: Calculate Interest

Interest = ₹8,00,000 × 10% = ₹80,000

Step 3: Apply Indifference Point Equation

At the Point of Indifference:

$$EPS_1 = EPS_2$$

$$EBIT(1 - T) / 1,00,000 = (EBIT - 80,000)(1 - T) / 60,000$$

Common Factor: Since (1 - T) is common on both sides, it is cancelled.

$$EBIT / 1,00,000 = (EBIT - 80,000) / 60,000$$

POINT OF INDIFFERENCE – PRACTICAL SOLUTION, ANSWER & DECISION

Cross Multiplication & Simplification

$$60,000 \text{ EBIT} = 1,00,000 (\text{EBIT} - 80,000)$$



$$0.6 \text{ EBIT} = \text{EBIT} - 80,000$$



$$0.4 \text{ EBIT} = 80,000$$



$$\text{EBIT} = ₹2,00,000$$

Answer

EBIT INDIFFERENCE POINT = ₹2,00,000

At an EBIT of ₹2,00,000, both the All-Equity Plan and Equity + Debt Plan will give the same EPS.

Decision Logic

Compare Actual EBIT with ₹2,00,000



EBIT < Indifference Point	EBIT = Indifference Point	EBIT > Indifference Point
All-equity plan may provide higher EPS because debt interest reduces earnings available to equity shareholders.	Both financing plans provide the same EPS.	Debt plan may provide higher EPS because fewer shares can outweigh the fixed interest burden.

Conclusion: Above ₹2,00,000, the debt plan may provide a higher EPS; below ₹2,00,000, the all-equity plan may be preferable from the EPS point of view. At ₹2,00,000, both plans provide equal EPS.

Final Quick Revision

Point of Indifference → EBIT where two financing plans give the same EPS

Basic Condition → $\text{EPS}_1 = \text{EPS}_2$

Above the Point → Debt may be better from EPS point of view

Below the Point → Equity may be better from EPS point of view

At the Point → Both plans give equal EPS